Operator Mapping and Homomorphism, Operation Laws

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1 Introduction

Consider operators (at least observables), which are abstract matrices in an undetermined representation, and abstract state vectors existing in a domain space D. There exists a map ϕ that maps elements from D to a codomain space C, where the image is the analytical representation of the preimage in a specific representation.

For example, the momentum operator $\hat{P}$ exists in D, and its image in the position representation is the element $-i\hbar\frac{d}{dx}$ in C.

2 Homomorphism Condition & Relations of Operation Laws

We now wish to discuss when a restricted map $\phi' : D' \to C'$ (where $D' \subset D$, $C' \subset C$, under the same mapping relation) is a homomorphism. For $d \in D'$, $c \in C'$, the homomorphism condition is:

$$\phi'(dc) = \phi'(d)\phi'(c)$$

This homomorphism can be achieved **if and only if the elements in C' are associative**.

- First, note that all elements in D satisfy associativity. However, their images in C do not necessarily satisfy it.
- Some elements in D commute. This is equivalent to their images in C commuting, which in turn is equivalent to those specific images being associative in C.
- For matrix elements in D, if two of them commute ($[\hat{A}, \hat{B}] = 0$), then there exists at least one basis in which both matrices are simultaneously diagonalized. Their images in C under this basis representation will commute.

3 Matrix & State Vector

Now, we attempt to incorporate state vectors into this description. If success, the theory is harmonic.

- **For a matrix element and a state vector element in** D, even in the basis where the matrix is diagonal, it is unclear how to define a "commutation relation" such that if the two elements commute in D, their images in C will also commute and be associative.

  (Example: If commutation is defined as $\hat{A}|\psi\rangle = (\langle\psi|\hat{A})^T$, then orthogonal matrices satisfy this, but orthogonal matrices are not necessarily diagonal. Therefore, their images in C do not necessarily commute with the image of the state vector.)

- **For two state vector elements in** D, one can define an operation between them (e.g., forming an outer product). The corresponding "commutation-like" relation for the resulting operators can be defined as:

  $$|\psi_1\rangle\langle\psi_2| = (|\psi_2\rangle\langle\psi_1|)^\dagger$$

  Taking the inner product induced by this operator equality implies:

  $$\langle\psi_1|\psi_2\rangle = \langle\psi_2|\psi_1\rangle^*$$

  This relation (conjugate symmetry of the inner product) holds universally in a complex Hilbert space. Within the *same representation*, the images of these state vectors and their outer products in C will satisfy the corresponding commutation and associativity rules under the standard function operations.

4 "Commutability"

Let's define a property called **"Commutability"**: Diagonal matrices and state vectors possess this property. Then, for an operation involving a sequence of elements from D:

- If **all elements possess this "Commutability"** property, they satisfy commutation, and the homomorphism ϕ' holds.
- If **any single element lacks this property**, commutation is not satisfied, and the homomorphism fails.

5 Mapping Properties

Simultaneously, for the map ϕ:

- If $\hat{A} \mapsto a$, then:

  - $\hat{A}^T \mapsto a^*$ (Transpose maps to Complex Conjugate)
  - $\hat{A}^\dagger \mapsto a$ (Hermitian Conjugate maps to the original function)

- If $|\psi\rangle \mapsto f$, then:
  - $|\psi\rangle^T \mapsto f$ (Transpose maps to the original function)
  - $|\psi\rangle^\dagger \mapsto f^*$ (Hermitian Conjugate maps to Complex Conjugate)

In summary:

- The **transpose** operation in D corresponds to **complex conjugation** in C.

- The **Hermitian conjugate** operation in D corresponds to **no operation** (identity map) in C for operators, but to **complex conjugation** for state vectors.

The transpose and Hermitian conjugate operations themselves are **not** included within the homomorphic operation. However, complex conjugation *is* involved in the homomorphic operation. The "homomorphic operation" here refers to the exponentiation within the mapping: $\phi(a^k) = [\phi(a)]^k$.